QUBO Formulation: Hospital Staff Rescheduling Problem

Hospital Staff Rescheduling Problem

Problem Statement. Let $\mathcal{S}$ denote the set of available staff members and $\mathcal{J}$ denote the set of uncovered shifts. Let $\mathcal{A} \subseteq \mathcal{S} \times \mathcal{J}$ be the set of qualified staff-shift pairs, where $(s, j) \in \mathcal{A}$ indicates that staff member s is qualified to cover shift j. For each qualified pair $(s, j) \in \mathcal{A}$, let $c_{s,j} \geq 0$ represent the cost of assigning staff s to shift j, which may encode fatigue, travel distance, overtime, or preference penalties. The objective is to select a set of assignments such that every shift in $\mathcal{J}$ is covered by exactly one staff member, each staff member covers at most one shift, and the total assignment cost is minimized.

Decision Variables. For each qualified pair $(s, j) \in \mathcal{A}$, introduce a binary decision variable $x_{s,j} \in \{0, 1\}$. The variable $x_{s,j}$ takes the value 1 if staff member s is assigned to shift j, and 0 otherwise. The qualification constraint is enforced by restricting the variable domain to $\mathcal{A}$; variables corresponding to unqualified pairs are excluded from the formulation.

QUBO Derivation. We construct a Quadratic Unconstrained Binary Optimization (QUBO) formulation by converting the constrained minimization problem into an unconstrained energy minimization problem $H(\mathbf{x})$.

Step 1: Objective Function. The original objective is to minimize the total assignment cost:

$$\min \sum_{(s,j)\in\mathcal{A}} c_{s,j} x_{s,j}. \tag{1}$$

This term is retained directly in the QUBO Hamiltonian.

Step 2: Constraints. The problem imposes two types of constraints:

1. *Shift Coverage:* Each shift $j \in \mathcal{J}$ must be assigned exactly one staff member. Let $\mathcal{S}_j = \{s \in \mathcal{S} : (s, j) \in \mathcal{A}\}$ be the set of qualified staff for shift j. The constraint is:

$$\sum_{s\in\mathcal{S}_j} x_{s,j} = 1 \quad \forall j \in \mathcal{J}. \tag{2}$$

2. *Staff Limit:* Each staff member $s \in \mathcal{S}$ can cover at most one shift. Let $\mathcal{J}_s = \{j \in \mathcal{J} : (s, j) \in \mathcal{A}\}$ be the set of shifts qualified for staff s. The constraint is:

$$\sum_{j \in \mathcal{J}_s} x_{s,j} \leq 1 \quad \forall s \in \mathcal{S}. \tag{3}$$

Step 3: Penalty Terms. We convert constraints to quadratic penalty terms using a penalty weight $\lambda > 0$.

Shift Coverage Penalty: For each shift j, we enforce equality using the squared penalty:

$$P_{\text{shift},j} = \lambda \left(\sum_{s \in \mathcal{S}_j} x_{s,j} - 1 \right)^2. \tag{4}$$

Expanding using the idempotent property $x_{s,j}^2 = x_{s,j}$ for binary variables:

$$P_{\text{shift},j} = \lambda \left(\sum_{s \in \mathcal{S}_j} x_{s,j}^2 - 2 \sum_{s \in \mathcal{S}_j} x_{s,j} + 1 \right) \tag{5}$$

$$= \lambda \left(- \sum_{s \in \mathcal{S}_j} x_{s,j} + 2 \sum_{\substack{s,s' \in \mathcal{S}_j \\ s<s'}} x_{s,j} x_{s',j} + 1 \right). \tag{6}$$

Summing over all shifts gives the total shift penalty $H_{\text{shift}} = \sum_{j \in \mathcal{J}} P_{\text{shift},j}$.

Staff Limit Penalty: For each staff member s, we enforce the inequality $\sum_{j \in \mathcal{J}_s} x_{s,j} \leq 1$ using the squared penalty $\lambda \left(\sum_{j \in \mathcal{J}_s} x_{s,j} \right)^2$, which is zero for sums 0 or 1 and positive for sums greater than 1:

$$P_{\text{staff},s} = \lambda \left(\sum_{j \in \mathcal{J}_s} x_{s,j} \right)^2. \tag{7}$$

Expanding:

$$P_{\text{staff},s} = \lambda \left(\sum_{j \in \mathcal{J}_s} x_{s,j}^2 + 2 \sum_{\substack{j,j' \in \mathcal{J}_s \\ j<j'}} x_{s,j} x_{s,j'} \right) \tag{8}$$

$$= \lambda \left(\sum_{j \in \mathcal{J}_s} x_{s,j} + 2 \sum_{\substack{j,j' \in \mathcal{J}_s \\ j<j'}} x_{s,j} x_{s,j'} \right). \tag{9}$$

Summing over all staff gives the total staff penalty $H_{\text{staff}} = \sum_{s \in \mathcal{S}} P_{\text{staff},s}$.

Step 4: Hamiltonian. The total QUBO Hamiltonian combines the objective and penalty terms. We subtract a constant offset to remove the constant terms arising from the shift coverage expansion:

$$H(\mathbf{x}) = \sum_{(s,j)\in\mathcal{A}} c_{s,j}x_{s,j} + H_{\text{shift}} + H_{\text{staff}} - \lambda|\mathcal{J}|. \tag{10}$$

Step 5: Q Matrix and Offset. Collecting coefficients for linear and quadratic terms yields the QUBO parameters. The Hamiltonian can be written as $H(\mathbf{x}) = \sum_{(s,j)\in\mathcal{A}} Q_{(s,j),(s,j)}x_{s,j} + \sum_{(s,j)\neq(s',j')\in\mathcal{A}^2} Q_{(s,j),(s',j')}x_{s,j}x_{s',j'} + c$. The entries are:

$$Q_{(s,j),(s,j)} = c_{s,j} - \lambda + \lambda = c_{s,j}, \tag{11}$$
$$Q_{(s,j),(s',j)} = 2\lambda \quad \text{for } s \neq s', \tag{12}$$
$$Q_{(s,j),(s,j')} = 2\lambda \quad \text{for } j \neq j', \tag{13}$$
$$Q_{(s,j),(s',j')} = 0 \quad \text{otherwise}, \tag{14}$$
$$c = 0. \tag{15}$$

Here, $Q_{(s,j),(s',j)}$ corresponds to pairs sharing the same shift, and $Q_{(s,j),(s,j')}$ corresponds to pairs sharing the same staff member.

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any violation of the constraints results in an energy increase that outweighs any potential reduction in the objective cost. Specifically, let $C_{\max} = \max_{(s,j)\in\mathcal{A}} c_{s,j}$. Choosing $\lambda > C_{\max}$ guarantees that the penalty for violating a constraint exceeds the maximum possible gain from the objective function. For the shift coverage constraint, a violation allows at most saving $C_{\max}$ in cost (by omitting an assignment), while incurring a penalty of at least λ. For the staff limit constraint, a violation incurs a penalty of at least λ with no cost reduction. Thus, $\lambda > C_{\max}$ ensures that infeasible solutions have strictly higher energy than the best feasible solution.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0,1\}^{|\mathcal{A}|}$ recovers the optimal assignment schedule. For any feasible solution, the penalty terms vanish (after offset subtraction), so $H(\mathbf{x})$ equals the total assignment cost. For any infeasible solution, at least one penalty term is positive, contributing at least λ to the energy. Since $\lambda > C_{\max}$, the energy of any infeasible solution exceeds the cost of any feasible solution (assuming costs are non-negative and at least one feasible assignment exists). Therefore, the global minimum of $H(\mathbf{x})$ must be a feasible solution with minimum total cost, corresponding to the optimal staff rescheduling.

Illustrative Example. Consider a small instance with $|\mathcal{S}| = 2$ staff members ($s \in \{0,1\}$) and $|\mathcal{J}| = 2$ shifts ($j \in \{0,1\}$), where all staff are qualified for all shifts, so $\mathcal{A} = \mathcal{S} \times \mathcal{J}$. Let the cost matrix be:

$$c_{0,0} = 10, \quad c_{0,1} = 20, \quad c_{1,0} = 15, \quad c_{1,1} = 5. \tag{16}$$

The decision variables are $x_{0,0}, x_{0,1}, x_{1,0}, x_{1,1}$. The QUBO Hamiltonian is:

$$\begin{aligned} H(\mathbf{x}) = {} & 10x_{0,0} + 20x_{0,1} + 15x_{1,0} + 5x_{1,1} \\ & + \lambda(x_{0,0} + x_{1,0} - 1)^2 + \lambda(x_{0,1} + x_{1,1} - 1)^2 \\ & + \lambda(x_{0,0} + x_{0,1})^2 + \lambda(x_{1,0} + x_{1,1})^2 - 2\lambda. \end{aligned} \tag{17}$$

Expanding and collecting terms yields the Q matrix entries consistent with the general derivation. For $\lambda > 20$, the optimal solution minimizes the cost subject to constraints. The feasible assignments are $(x_{0,0} = 1, x_{1,1} = 1)$ with cost 15, and $(x_{0,1} = 1, x_{1,0} = 1)$ with cost 35. The global minimum is achieved at:

$$x_{0,0} = 1, \quad x_{1,1} = 1, \quad x_{0,1} = 0, \quad x_{1,0} = 0, \tag{18}$$

with minimum energy $H(\mathbf{x}^*) = 15$. This corresponds to assigning staff 0 to shift 0 and staff 1 to shift 1, yielding the minimum total cost of 15.